



PowerFactory 2024

DIGSILENT Doubly Fed Induction Generator Templates

PF2024

POWER SYSTEM SOLUTIONS
MADE IN GERMANY

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1 Introduction

This document describes the DFIG template “DIgSILENT DFIG WTG 2.5MW” that is delivered with *PowerFactory*, its applicability and possible configurations.

The “DIgSILENT DFIG WTG 2.5MW” template is a generic model of a wind turbine with a doubly fed induction generator. The model represents one wind turbine (the number of parallel machines can be changed as explained in this document) and accounts for the PQ characteristic and short-circuit contribution of the generator (steady-state analysis), the dynamic controllers of the DFIG and converters as well as the mechanical part of the rotor and the aerodynamics.

The DFIG model could be used for:

- Load Flow Studies
- Stability Studies (RMS)
- Transient Studies (EMT)
- Balanced and unbalanced simulations (however control is implemented for the positive sequence only)

The following DFIG model for 50 Hz and 60 Hz applications are available in the global template library:

- DIgSILENT DFIG WTG 2.5MW 50Hz
- DIgSILENT DFIG WTG 2.5MW 60Hz

Each template contains also three transformer types for 10, 20 and 30 kV. The model from the template is at the beginning equipped with the 20 kV transformer.

2 How to use this Template

PowerFactory contains pre-defined templates of dynamic models for different generic wind turbines, photovoltaic generators and a battery energy storing system. Using the templates is simple, just click on the **General Templates** button () and select the desired template in the following window. Thereafter you can place the model in the single line diagram. For this you have to click one time in the single line diagram, this shows the model connected to the mouse pointer a second click creates a new model of the selected template in the single line diagram. Each dynamic model consists of some elements in the single line diagram (mostly a terminal with a generator) plus a composite model, which is located in the grid or the recording variation.

After you placed all models in the single line diagram you have to close the selection window with the templates, this terminates the template inserting.

Do not delete the terminal to which the generator is connected - this would destroy the measurement points of all measurement devices (located in the corresponding composite model).

You could delete a placed model directly by pressing the undo button () or you could delete the model also manually. If you want to delete a model manually you have to mind that the model consists of visible elements in the single line diagram and also of a composite model which is only visible in the Data Manager.

3 Model Description

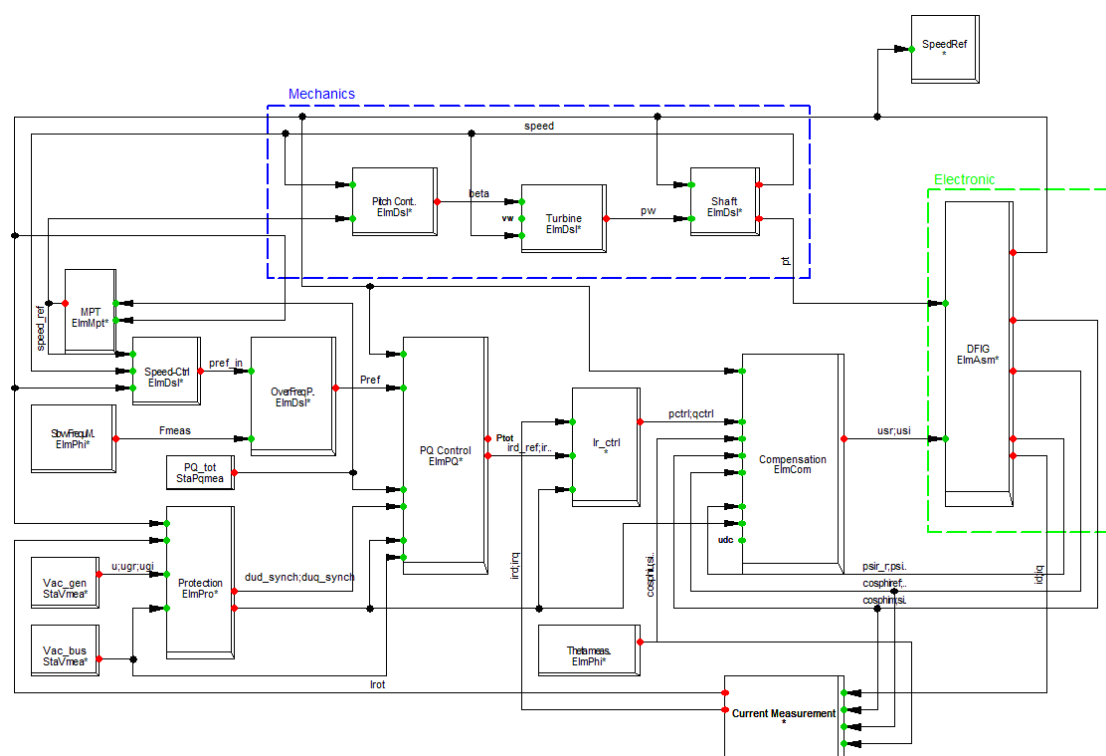
The doubly fed induction generator (DFIG) is in the single line diagram represented by an asynchronous machine which is configured as a DFIG. The model of the controller and the dynamic parts are collected in the composite model *DFIG Control*. This composite model can be found either through the link *Plant Model* on the Basic Data page of the DFIG or with the Data Manger in the used grid.

The composite model *DFIG Control* is created from the frame definition *Generic DFIG- Turbine_resync*. The graphical definition of this frame is shown in Figure 3.1.

The single slots are briefly described in Table 3.1. Each slot could be filled with either a DSL-model, a measurement device or a *PowerFactory* element, such as a asynchronous machine. All measurement devices are connected either to the terminal or to the cubicle, which connects the generator with the terminal. This approach ensures that all measurement devices are correctly connected after entering the model using the template.

Table 3.1: Frame description

Slot Name	Description	Needed Type
Compensation	This models calculates a coordinate transformation of the rotor voltage	DSL-Model
Current Measurement	Calculates the rotor current from angle and id/iq of the DFIG	DSL-Model
DFIG	Asynchronous machine, configured as DFIG	*.ElmAsm
Ir-ctrl	Calculates a rotor reference voltage for a given rotor current set point	DSL-Model
MPT	Maximum power tracker, calculates optimal speed for max. power	DSL-Model
OverFreq Pwr Reduction	Reduces the power in case of electrical over frequency	DSL-Model
Pitch Control	Controls the pitch angle of the rotor	DSL-Model
PQ Control	Controls active and reactive power through the rotor current	DSL-Model
PQ_tot	Active and reactive power measurement device	*.StaPqmea
Protection	Triggers bypass, in case of high rotor current, high speed or over voltage Could also disconnect and resynchronize the DFIG	DSL-Model
Shaft	Calculates from wind power and generator speed the mechanical power and the rotor speed	DSL-Model
SlowFrequMeas	Slow frequency measurement device (PLL)	*.ElmPhi
Speed-Ctrl	Calculates a reference power from speed and reference speed	DSL-Model
SpeedRef	Reference speed, needed only for initialisation of the mechanical generator speed	DSL-Model
Theata meas.	Fast frequency and voltage angle measurement device (PLL)	*.ElmPhi
Turbine	Calculates from pitch angle, rotor speed and wind speed the wind power	DSL-Model
Vac.bus	AC-voltage measurement device on bus side	*.StaVmea
Vac.gen	AC-voltage measurement device on generator side	*.StaVmea

Figure 3.1: Frame Definition *Generic DFIG-Turbine_resync* (*.BlkDef)

4 How to Customize the Model

You will find in this chapter a description on how you can configure the model from the template according to your specific needs. The configuration is sorted to the calculation functions because there are fewer configurations needed for load flow calculation than for a dynamic simulation.

4.1 Load Flow

Changing the number of parallel machines

For grid impact studies of a wind farm it is often sufficient to model the wind farm with one aggregated machine. The generator from the template represents only one machine, but this could be changed very fast in the generator element on the basic data page. For load flow studies are no further changes needed.

You have to change also the number of parallel transformers if you added a step up transformer to the wind turbine.

Changing the nominal power

The nominal power is for the load flow model only defined in the type of the generator. If you want to change the nominal power for all the machines of the template you could change the nominal power of the type which is stored in the project Library\Templates\DIgSILENT DFIG WTG 2.5MW\Library\Asynchronous Machine. If you want to change the nominal power for only

one machine you have to copy&paste the type and change then this type.

Changing the nominal AC voltage

You could also change the nominal voltage of the machine if your machine has a different nominal rotor voltage. If you want to change the nominal voltage for all the machines of the template you could change the nominal voltage of the type which is stored in the project Library\Templates\DlgSILENT DFIG WTG 2.5MW\Library\Asynchronous Machine. If you want to change the nominal power for only one machine you have to copy&paste the type and change then this type. Mind that you have also to change the nominal voltage of the terminal of the asynchronous machine.

4.2 Short Circuit (VDE / IEC / ANSI / complete)

The short circuit behaviour of the model is also completely determined by the element and type settings. There are therefore no changes in the dynamic model parts necessary. It is important to note that the standard short circuit methods could not give precise results for a controlled wind turbine. For more precise results are dynamic simulations needed. The short circuit results of the DFIG are determined by the used asynchronous machine type. Please read the technical document for the asynchronous machine for more details [3].

4.3 Dynamic Simulation (RMS/EMT)

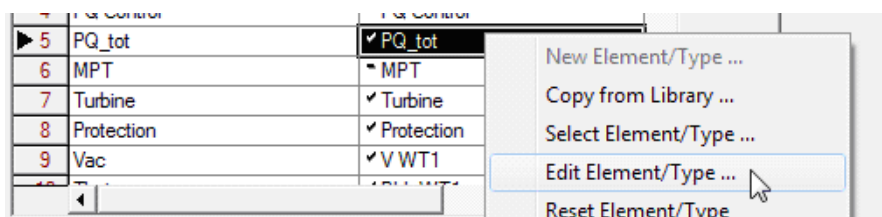
The main purpose of the model created with the template is of course the dynamic simulation. The great advantage of using a template for creating the wind turbine model is that all measurement points are already correct defined and all common models are already configured.

The model is generic and therefore completely open - you could configure it according to your needs. Following you will find a short manual for changing the number of parallel machines, for changing the rated power and for changing some protection settings.

Changing the number of parallel machines

To change the number of parallel machines you have to follow the next two steps:

1. Change the number of parallel machines as described for the load flow (Chapter 4.1).
2. Open the composite model *DFIG Control*. To find the correct composite model (if you have more than one) you could double click on the DFIG in the single line diagram. On the basic data page is a link to the composite model (Plant Model). In the composite model you have to open (double click or right click and *Edit Element/Type...*) the power measurement device *PQ_tot*:



The rated power has to be set to:

$$P_{rating} = n_{parallel} \cdot P_{DFIGrated}$$

Example: 10 parallel machines and a rated active power of 2 MW per machine results in 20 MW.

If you added a step-up transformer you have to change here also the number of parallel devices.

Changing the rated power of the model

Please check if there is already a DFIG model with the needed rated power in the template library before you change the rated power of the DFIG model by hand. If there is no fitting template you have to follow the steps below.

To change the rated power of the DFIG to simulate for example a 3 MW DFIG instead of the given 2.5 MW you have to follow step 1 to 7. You should have also the amount of parallel machines in mind. The rated power of the DFIG is stored in the type data. It is recommended to generate a new type for the DFIG with the new rated power.

1. Open the Data Manager and switch in the Library\Templates\DlgSILENT DFIG WTG 2.5MW\Library\Asynchronous Machine
2. Copy and paste the asynchronous machine type.
3. Rename the copied type.
4. Open the type and configure it according to your needs:

Asynchronous Machine Type - Library\DFIG Library\Asynchronous Machines\IG 3MW.TypeAsmo

ANSI Short-Circuit | IEC 61363 | RMS-Simulation | EMT-Simulation | Harmonics | P
Optimization | State Estimator | Reliability | Descript
Basic Data | Load Flow | VDE/IEC Short-Circuit | Complete Short-C

Name: IG 3MW

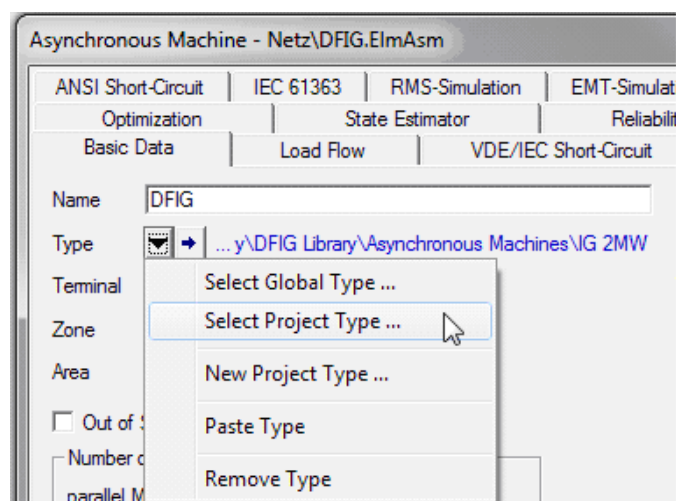
Rated Voltage: 0.69 kV

Input Mode:
☐ Slip-Torque/Current Characteristic
☒ Electrical Parameter

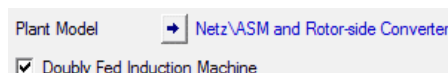
Power Rating:
☐ Rated Apparent Power: 3422.901 kVA
☒ Rated Mechanical Power: 3000. kW

(In the example above the rated mechanical power is changed to 3 MW)

5. Select the new type in the DFIG:



6. Switch to the Load Flow page and adjust also the active and reactive power.
7. Open the composite model *DFIG Control*. To find the correct composite model (if you have more than one) you could double click on the DFIG in the single line diagram. On the basic data page is a link to the composite model:



Here you have to change:

- The *Current Measurement*; open the element and change the apparent power (3422 kVA)
- The power measurement *PQ_tot*; here you have to enter the active power of the DFIG times the number of parallel machines. In case of 30 parallel machines and 3 MW you have to enter here 90 MW

Please not that you have also to match the used transformer type.

4.3.1 Changing the Protection Settings of the DFIG

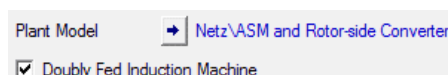
The protection of the DFIG consists of a crowbar, the possibility to disconnect the DFIG and also the possibility to re-synchronise it. If the speed, the voltage or the rotor current exceeds a predefined limit for a certain time the DFIG will be disconnected from the busbar. In case of an over current (this could be caused by a short circuit) an additional inductance is inserted in the rotor circuit of the DFIG. This additional inductance is called crowbar. The standard settings for the protection are given in the table below:

Table 4.1: Protection Block Settings

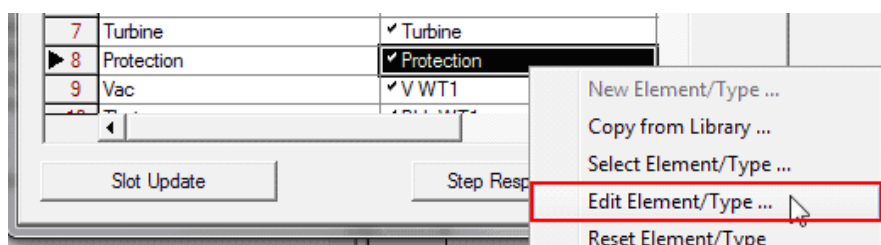
Parameter	Description	Value	Unit
Maxirotor	Rotor Current for Crowbar Insertion	2,0	pu
tbypass	Crowbar Insertion Time	0,06	s
TripIfCrow	Disconnect the DFIG if the Crowbar is inserted	1	[0/1]
MaxSynch	Time Duration of the synchronisation window	2,0	s
CloseDelay	Breaker delay after synchronisation	0,05	s
Usynch	Minimum grid voltage for reconnection	0,9	pu
Reconnect	Reconnect after trip	1	[0/1]
MaxSpeed1	Overspeed Setting step 1	1,5	pu
ttripMaxS1	Overspeed Time Setting step 1	0	s
MaxSpeed2	Overspeed Setting step 2	1,4	pu
ttripMaxS2	Overspeed Time Setting step 2	1,0	s
MinSpeed1	Underspeed Setting step 1	0,6	pu
ttripMinS1	Underspeed Time Setting step 1	0	s
MinSpeed2	Underspeed Setting step 2	0,7	pu
ttripMinS2	Underspeed Time Setting step 2	1,0	s
MaxVoltage1	Overvoltage Setting step 1	1,5	pu
ttripMaxV1	Overvoltage Time Setting step 1	0,1	s
MaxVoltage2	Overvoltage Setting step 2	1,2	pu
ttripMaxV2	Overvoltage Time Setting step 2	1,0	s
MinVoltage1	Undervoltage Setting step 1	0,2	pu
ttripMinV1	Undervoltage Time Setting step 1	1,0	s
MinVoltage2	Undervoltage Setting step 2	0,4	pu
ttripMinV2	Undervoltage Time Setting step 2	2,0	s

If you want to change the protections settings shown above you have to follow step 1 and 2:

1. Open the composite model *ASM and Rotor-side Converter*. To find the correct composite model (if you have more than one) you could double click on the DFIG in the single line diagram. On the basic data page is a link to the composite model:



2. Double click or right click and select *Edit Element/Type...*:



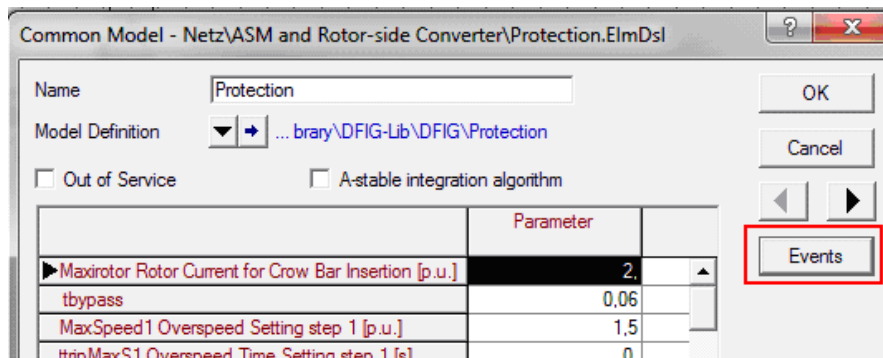
In the following dialogue you could alter the protection values.

Changing the inductance of the crowbar

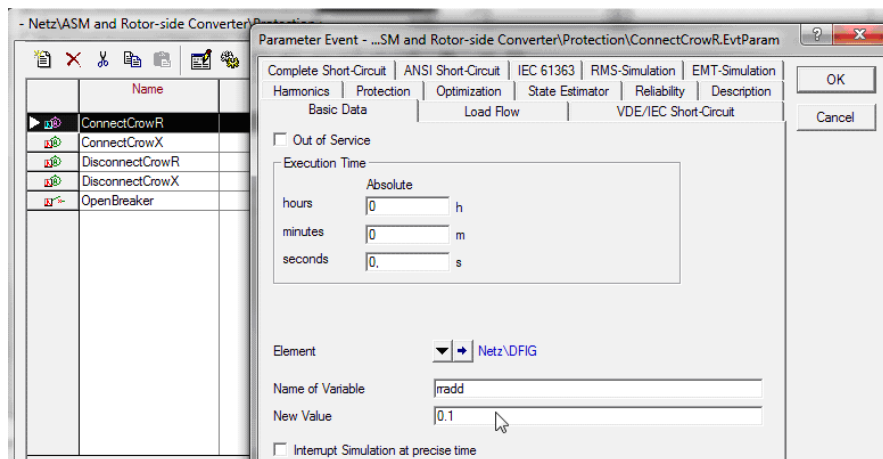
For changing the inductance of the crowbar you have to alter the resistance and the reactance in the parameter events. Follow step 1 and 2 above and when:

4 How to Customize the Model

1. Click on **Events**:



2. In the following dialogue you see five events. Two events that connects the crowbar, two events that disconnects the crowbar and one open breaker event. Double click on the *ConnectCrowR* event to change the resistance of the crowbar (New Value). Double click on the *ConnectCrowX* event to change the reactance the crowbar:



The resistance and inductance are per unit values on the basis of the asynchronous machine type.

The crowbar triggers too fast

If the crowbar is triggered too fast you could change the rotor current value *MaxIrotor*. The crowbar is only inserted if the rotor current exceeds the value of *MaxIrotor*.

The DFIG should not disconnect if the crowbar is inserted

The standard setting of the model is that the DFIG disconnects from the grid if the crowbar was triggered. If this is not desired the parameter *TriplIfCrow* could be changed to 0. This disables the triggering of the *OpenBreaker* event.

The DFIG should not reconnect after tripping

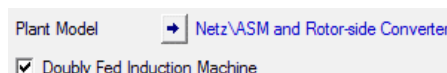
If the DFIG is disconnected from the grid and a re-synchronisation is not desired then the parameter *Reconnect* could be change to zero. This prevents the re-synchronisation.

Active power increase after resynchronisation is too fast/slow

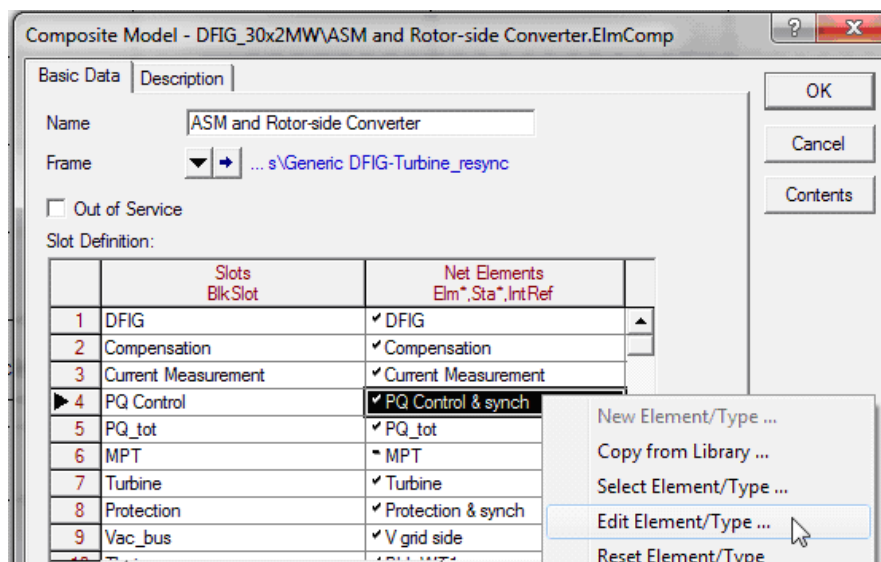
The gradient of the active power feed-in could be configured with the parameter *Tramp* in the *PQ Control & synch* block. This option is only active if *i_LEEG* is set to 0, if *i_LEEG* is 1 the machine behaves according to the Ordinance on System Services by Wind Energy Plants (SDLWindV) [2]. "The active power feed-in must be increased with a gradient of at least 10 % of the nominal generator capacity per second to the original value" according to the German Transmission Code 2007 [1].

If you want to change the protections settings shown above you have to follow step 1 to 3:

1. Open the composite model *ASM and Rotor-side Converter*. To find the correct composite model (if you have more than one) you could double click on the DFIG in the single line diagram. On the basic data page is a link to the composite model:



2. Double click or right click and select *Edit Element/Type...*:

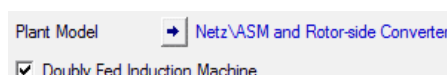


3. Change the parameter *Tramp* according to your needs.

Note: The unit is s/p.u. so a gradient of 20 % per second results in 5 s/p.u.

Changing active power reduction during over frequency

The WTG could also reduce the active power during over frequency conditions (for this is also the slow PLL needed). The behaviour of this active power reduction could be configured in the block *OverFrequPowerReduction*. This common model is part of the composite model *DFIG Control*. To find the correct composite model (if you have more than one) you could double click on the static generator in the single line diagram. On the basic data page is a link to the composite model:



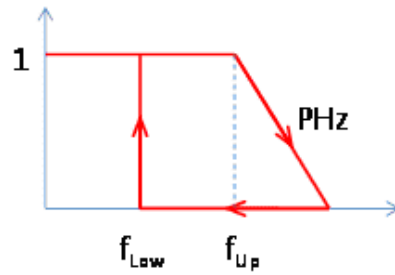


Figure 4.1: Active Power Reduction during Over Frequency

The implemented characteristic is shown in Figure 4.1. There are six parameters which could be changed:

Table 4.2: Parameters Active Power Reduction during Over Frequency

Parameter	Description
fUp	Frequency which triggers the active power reduction (higher than fLow)
fLow	Frequency which ends the active power reduction
PHz	Gradient of the active power reduction (proportional to over frequency)
Tfilter	PT1 filter for frequency measurement
negGrad	Gradient limitation for active power reduction
posGrad	Gradient limitation for active power increase

5 References

- [1] Transmission Code 2007, 2007.
- [2] Ordinance on System Services by Wind Energy Plants SDLWindV, 2009.
- [3] *Technical Reference Induction Machine*. Heinrich-Hertz-Strasse 9, 72810 Gomaringen / Germany, 2010.

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